leetcode.com/problems/maximum-subarray/

Problem List

Submit

Premium

Description | Editorial | Solutions | Submissions

53. Maximum Subarray

Medium | Topics | Companies

Given an integer array `nums`, find the subarray with the largest sum, and return *its sum*.

Example 1:

> **Input:** nums = [-2,1,-3,4,-1,2,1,-5,4]
> **Output:** 6
> **Explanation:** The subarray [4,-1,2,1] has the largest sum 6.

Example 2:

> **Input:** nums = [1]
> **Output:** 1
> **Explanation:** The subarray [1] has the largest sum 1.

Example 3:

> **Input:** nums = [5,4,-1,7,8]
> **Output:** 23
> **Explanation:** The subarray [5,4,-1,7,8] has the largest sum 23.

Constraints:

- $1 <= nums.length <= 10^5$
- $-10^4 <= nums[i] <= 10^4$

Follow up: If you have figured out the `O(n)` solution, try coding another solution using the **divide and conquer** approach, which is more subtle.

38.3K | 583 | 434 Online

Code

Java | Auto

```java
class Solution {
    public int maxSubArray(int[] nums) {
        int currentSum = nums[0];
        int maxSum = nums[0];

        for (int i = 1; i < nums.length; i++) {
            currentSum = Math.max(nums[i], currentSum + nums[i]);

            maxSum = Math.max(maxSum, currentSum);
        }

        return maxSum;
    }
}

```

Saved | Ln 15, Col 1

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2 | Case 3

Input

nums =

[-2,1,-3,4,-1,2,1,-5,4]

Output

6

Expected

6

Contribute a testcase

leetcode.com/problems/find-the-highest-altitude/

Problem List

Submit

Premium

Description | Editorial | Solutions | Submissions

1732. Find the Highest Altitude

Easy | Topics | Companies | Hint

There is a biker going on a road trip. The road trip consists of `n + 1` points at various altitudes. The biker starts his trip on point `0` with altitude equal `0`.

You are given an integer array `gain` of length `n` where `gain[i]` is the **net gain in altitude** between points `i` and `i + 1` for all (`0 <= i < n`). Return *the* ***highest altitude*** *of a point.*

Example 1:

```
Input: gain = [-5,1,5,0,-7]
Output: 1
Explanation: The altitudes are [0,-5,-4,1,1,-6]. The highest is 1.
```

Example 2:

```
Input: gain = [-4,-3,-2,-1,4,3,2]
Output: 0
Explanation: The altitudes are [0,-4,-7,-9,-10,-6,-3,-1]. The highest is 0.
```

Constraints:

- `n == gain.length`
- `1 <= n <= 100`
- `-100 <= gain[i] <= 100`

Discover more | Android Apps | Geographic Reference

Seen this question in a real interview before? 1/6

3.5K | 429 | 21 Online

Code

Java | Auto

```java
class Solution {
    public int largestAltitude(int[] gain) {
        int currentAltitude = 0;
        int highestAltitude = 0;

        for (int g : gain) {
            currentAltitude += g;

            highestAltitude = Math.max(highestAltitude, currentAltitude);
        }

        return highestAltitude;
    }
}

```

Saved | Ln 15, Col 1

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2

Input

gain =

[-5,1,5,0,-7]

Output

1

Expected

1

Contribute a testcase